Observational Coordinates and Estimates

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. Two coordinate systems, one sky

Catalogues use equatorial coordinates: right ascension α and declination δ. Observers use horizontal coordinates: altitude h and azimuth A. Altitude is measured above the horizon; azimuth is measured from north through east. The bridge between them is the hour angle

$$H = \mathrm{LST} - \alpha, \tag{1}$$

where LST is local sidereal time. $H = 0$ means the object is on the local meridian.

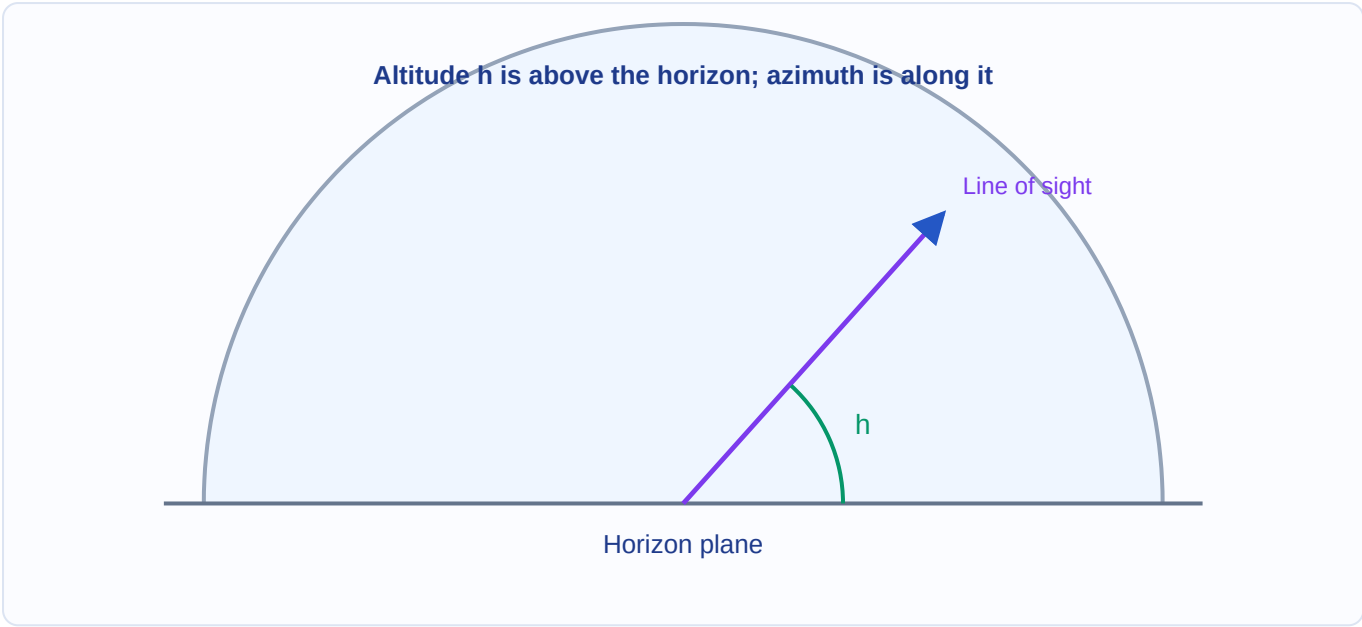


Figure 1. Hour angle links the catalogue grid to the local meridian.

2. Altitude and azimuth

For latitude ϕ, the altitude is

$$\sin h = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H. \tag{2}$$

Azimuth can be found from

$$\tan A = \frac{-\cos\delta \sin H}{\sin\delta \cos\phi - \cos\delta \cos H \sin\phi}, \tag{3}$$

with the quadrant chosen correctly. In practice, use an atan2 function or a spherical-astronomy diagram. At upper culmination, $H = 0$ and the altitude is $h = 90^\circ - |\phi - \delta|$.

3. Rise, set and circumpolar stars

An object rises or sets when $h = 0$. The hour angle at the horizon satisfies

$$\cos H_0 = -\tan\phi \tan\delta. \tag{4}$$

If $\cos H_0 < -1$, the object is circumpolar and never sets. If $\cos H_0 > 1$, it never rises. Otherwise it is above the horizon for $2H_0/15$ hours, with H_0 in degrees.

At latitude 48° N, stars with $\delta > 42^\circ$ are circumpolar; stars with $\delta < -42^\circ$ never rise.

4. Airmass and extinction

The zenith distance is $z = 90^\circ - h$. For moderate zenith distances, airmass is approximately

$$X \approx \sec z = \frac{1}{\sin h}. \tag{5}$$

At $h = 30^\circ$, $X = 2$. Near the horizon the plane-parallel approximation fails: $\sec z$ diverges and overestimates the finite atmospheric path. A common empirical fit is the Kasten–Young (1989) formula

$$X = \frac{1}{\sin h + 0.50576\,(h + 6.07995)^{-1.6364}}, \tag{6}$$

with h in degrees. It gives about 38 at the horizon, but real extinction also depends on aerosols, water vapour, curvature and the observer's altitude. Extinction increases with airmass, so observations are best near the meridian and at high altitude.

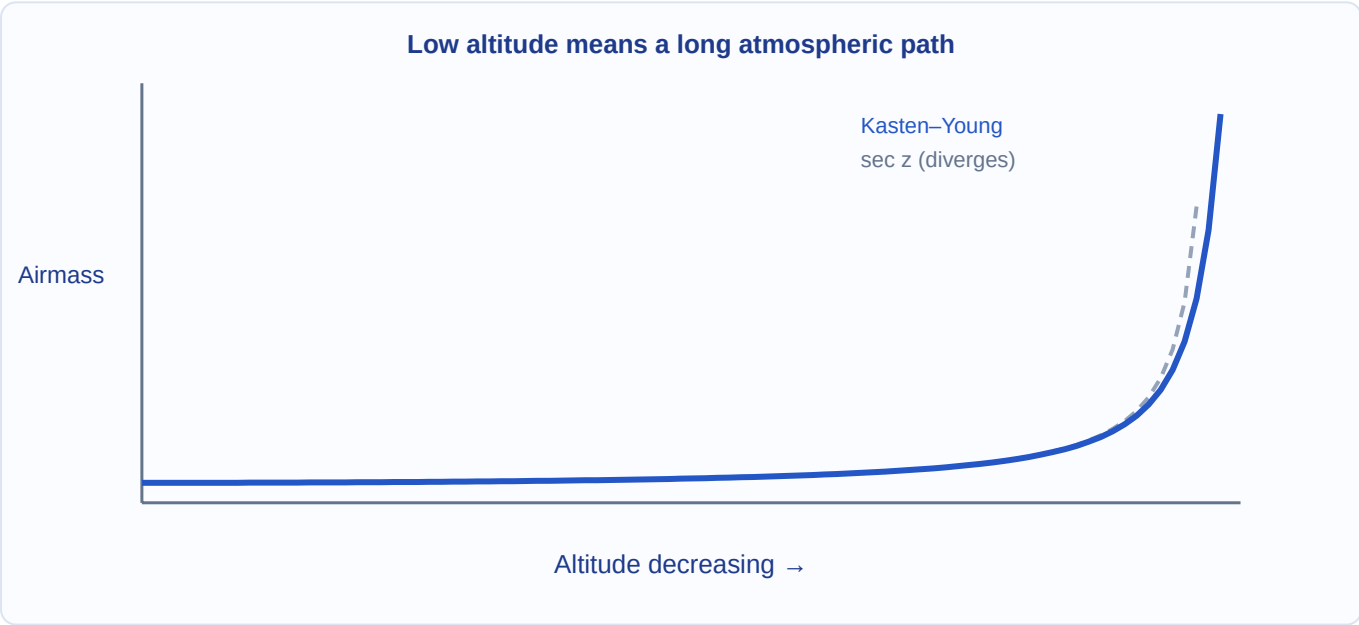


Figure 2. The plane-parallel sec z approximation overestimates the finite airmass near the horizon; the empirical curve remains finite.

5. Sidereal time and planning

Sidereal time tracks the stars, not the Sun. A sidereal day is about 23 h 56 m of solar time. Stars rise about 4 minutes earlier each solar day. To plan an observation, compute the LST at the desired time, find the hour angle, then compute altitude and airmass. Choose a window when the target is high and the Moon is far away.

6. Estimating in the field

If you know a bright star’s RA and Dec, you can estimate the LST from its hour angle. If you know the latitude, Polaris gives a rough north reference and its altitude approximates the latitude. Hand angles and a watch can turn a catalogue position into a practical pointing estimate.

INTERACTIVE EXPERIMENT

From catalogue coordinates to the local sky.

Hour angle H = LST − RA. Altitude and azimuth follow from spherical astronomy. Azimuth is measured from north through east.

Hour angle	Altitude	Azimuth
0.00°	**62.00°**	**180.00°**

Airmass	Rise/set state
1.13	**Above horizon**



Altitude and azimuth change with time and location; RA and Dec are catalogue coordinates. Airmass grows rapidly near the horizon; the lab uses the finite Kasten–Young approximation.

Try this. Set H = 0. Where is the object? Move the latitude and watch the circumpolar boundary change.

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Convert RA to degrees or hours consistently. Compute hour angle first. Then altitude, then azimuth. Check the meridian and horizon limits before trusting a number.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Hour angle. A star has RA 6 h. The LST is 8 h. Find the hour angle in hours and degrees. Is the star east or west of the meridian?

Worked solution →

[2] Problem 2 — Meridian altitude. At latitude 48° N, find the meridian altitude of a star with declination +20°.

Worked solution →

[2] Problem 3 — Altitude at hour angle. At latitude 48° N, a star has declination +20° and hour angle 30°. Estimate its altitude.

Worked solution →

[1] Problem 4 — Airmass. Find the approximate airmass at altitude 30° and at altitude 60°.

Worked solution →

[2] Problem 5 — Circumpolar limit. At latitude 48° N, what declination limit separates circumpolar stars from stars that set?

Worked solution →

[2] Problem 6 — Time above horizon. At latitude 48° N, a star has declination +20°. Find $\cos H_0$ and the approximate time above the horizon.

Worked solution →

[1] Problem 7 — Zenith distance. A star is at altitude 70°. Find its zenith distance and approximate airmass.

Worked solution →

[2] Problem 8 — Sidereal drift. A star transits at 22:00 tonight. Approximately what time will it transit 30 days later? Why?

Worked solution →

[3] Problem 9 — Azimuth at culmination. At latitude 48° N, a star with declination +20° transits. Is it due south or due north? What is its azimuth? What if the declination were +60°?

Worked solution →

[3] Problem 10 — Observing window. A target has RA 18 h and declination +30°. At a site at latitude 48° N, the LST is 20 h. Find the hour angle, altitude, azimuth and airmass. Is the target well placed?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.